Unit No. 4- Memory Organization

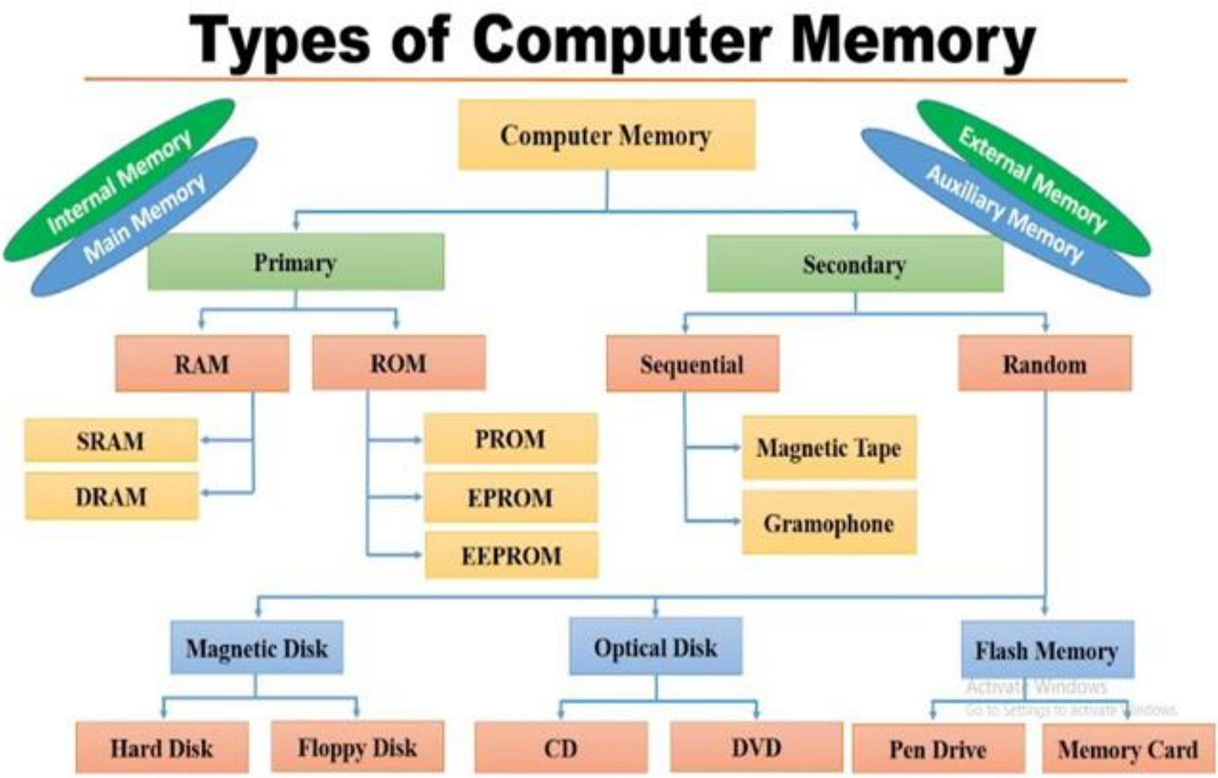
SEMICONDUCTOR MAIN MEMORY (INTERNAL MEMORY)

1. Introduction

Semiconductor main memory, also called **internal memory or primary memory**, is the memory unit directly accessible by the CPU. It is built using semiconductor integrated circuits (ICs) and stores:

- Program instructions
- Input data
- Intermediate results
- Final output before transfer to secondary storage

It acts as a working memory for the computer system.



Characteristics of Semiconductor Main Memory

1. **High Speed** – Faster than secondary storage.
2. **Direct CPU Access** – CPU can directly read/write.
3. **Limited Capacity** – Smaller than hard disks/SSDs.
4. **Costly per Bit** – More expensive than secondary memory.
5. **Mostly Volatile** – Data lost when power is off (except ROM).

Classification of Semiconductor Memory

Semiconductor main memory is broadly classified into:

1. **RAM (Random Access Memory)**
2. **ROM (Read Only Memory)**

1. **Random Access Memory (RAM)**

RAM is volatile memory. It allows both read and write operations.

Types of RAM:

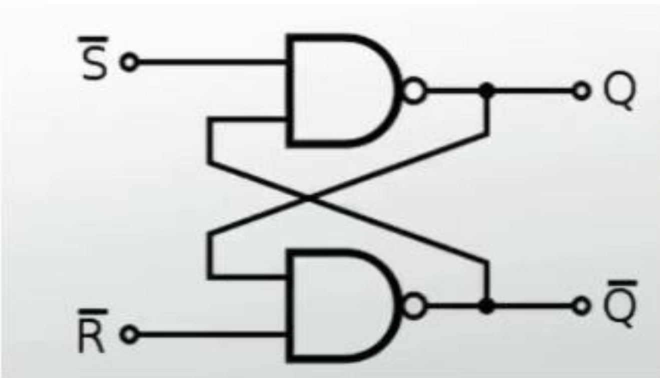
A. Static RAM (SRAM)

B. Dynamic RAM (DRAM)

A. SRAM (Static Random Access Memory)

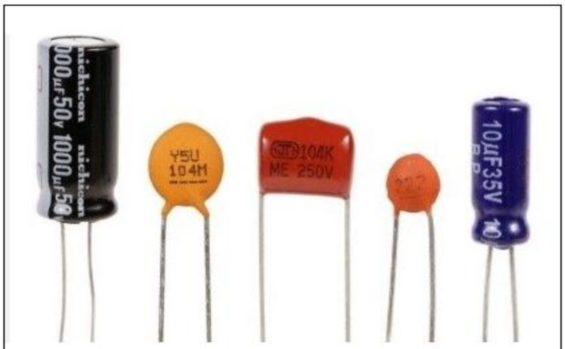
1. Introduction

- SRAM stands for **Static Random Access Memory**.
- It is the internal memory of the CPU for storing data, program, and program result. It is a read/write memory which stores data until the computer is working. As soon as the computer is switched off, data is erased. Therefore, RAM is a volatile memory.
- It is called **"Static"** because it can hold data as long as power is supplied, without needing periodic refreshing.
- Each memory cell of SRAM is made up of a flip-flop a 1-bit storage device. SRAM uses a 6 transistors. In this memory circuit, capacitors are not used. Thus, in SRAM, there is no data leakage, so SRAM need not be refreshed regularly.
- SRAM is mainly used in **cache memory** due to its very high speed.
- An array of flip-flops that are used to save the data. The memory cells consist of flip flops that hold the information till the power supply is on.
- Static RAM's are more expensive and consume more power and also have higher speeds than D-RAMs. Static RAM is used to build the CPU's speed-sensitive cache, while dynamic RAM forms the larger system RAM space.

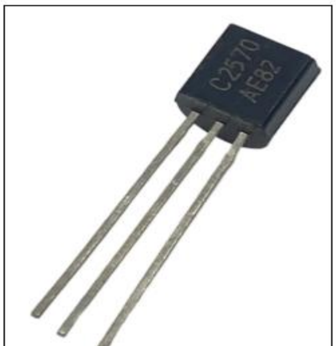


B. **Dynamic RAM (DRAM)**

- DRAM **(Dynamic Random Access Memory)**
- Each memory cell of DRAM is made up of one **transistor** and one **capacitor**.
- In DRAM, the data and information is stored in the form of an electric charged on the capacitor.
- Since capacitor loses its data (charge), thus DRAM must be continually refreshed several hundred times per second to maintain the data.
- It been wasted less power as compared to SRAM and operates at a slower rate than as well.
- DRAM is a small sized and less expensive
- It consists of memory cells, which constitute a single capacitor and a single transistor.



Capacitors



Transistor

What is Error Detection?

Error detection is the technique used to **identify whether data has been changed or corrupted** while it is being **transmitted** from one component to another or **stored** in memory/storage devices. To do this, the system adds **extra (redundant) bits** to the original data. At the receiving end, these bits are checked to see if the data is still correct.

If the received data does not match the expected pattern, an **error is detected**.

Where Error Detection Is Needed

1. **During Data Transmission**
2. **During Data Storage**

1.During Data Transmission

- Between **CPU and main memory**
- Over **system buses**
- Between **I/O devices and processor**
- Over **computer networks**

Example (Transmission Error): Suppose the CPU sends an 8-bit data value to memory
Original data sent: 10110101
Data received: 10110001

Here, the 5th bit has changed from 1 to 0.
Without error detection, memory would store **incorrect data**, leading to wrong program execution

2. During Data Storage

Errors can also occur **after data is stored**, especially in:

- RAM
- Cache memory
- Hard disks / SSDs

Example (Storage Error) -A value is stored in RAM:
Stored data: 11001010

Corrupted data: 11001110

This change is invisible unless an **error detection mechanism** checks the data.

Causes of Errors

1. Electrical Noise:
Electrical noise refers to **unwanted disturbances in electrical signals** that travel through circuits, wires, or buses in a computer system. These disturbances can **change the voltage level of a signal**, causing bits to be read incorrectly as `0` or `1`.
In digital systems, data is represented using voltage levels:

- **Logic 1** → High voltage
- **Logic 0** → Low voltage

When noise alters these voltage levels, **bit errors occur**.

2. Sudden Voltage Fluctuations:

Voltage may fluctuate due to: Sudden load changes in circuits

These fluctuations may push the signal voltage **above or below the threshold**, causing misinterpretation.

3.Crosstalk in Wires

Crosstalk occurs when signals in one wire **interfere with adjacent wires**, especially in:

- High-speed data buses
- Parallel communication lines

This happens because changing electric or magnetic fields from one wire induce unwanted signals in nearby

Types of Errors

1. Single-bit error: Only one bit is changed

Example: 1011001 → 1010001

Characteristics

- Most common in **simple transmission systems**
- Easy to detect using **parity check**
- Can be corrected using **Hamming code**

2.Multiple-bit error: More than one bit changes but not necessarily in sequence**.**

Characteristics

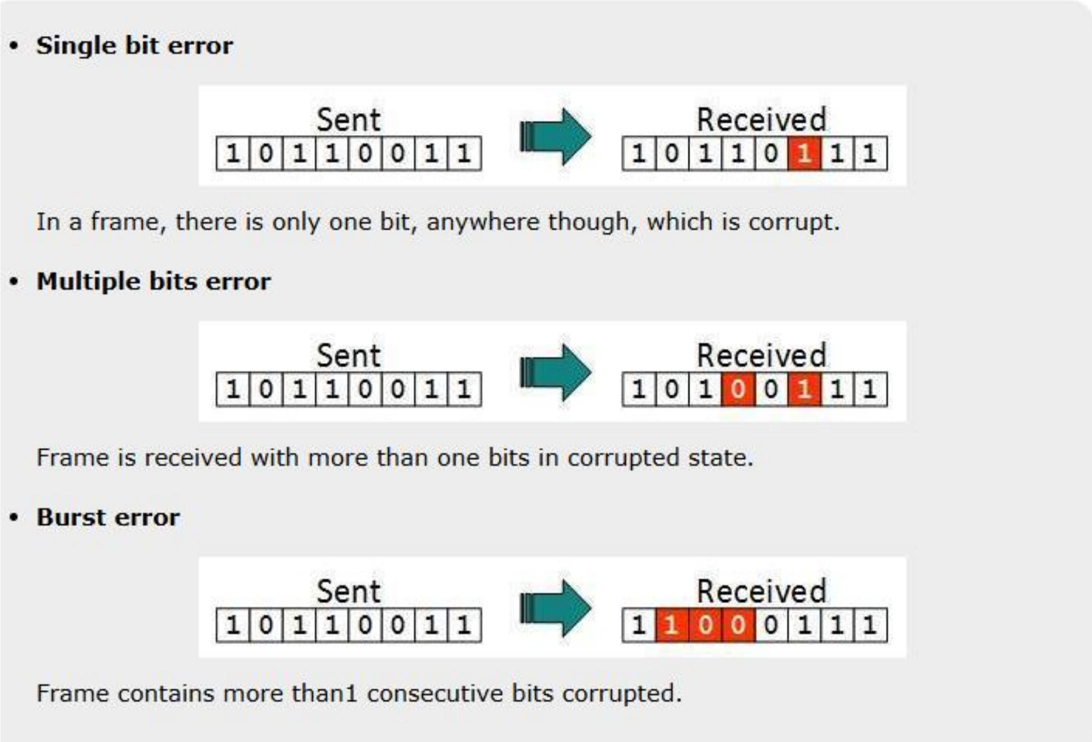
- Harder to detect than single-bit errors
- Simple parity check may **fail to detect** this error
- Requires stronger detection methods like **CRC-**Cyclic Redundancy Check

3.Burst error: A sequence of bits is affected.

The length of the burst is measured from the **first corrupted bit to the last corrupted bit**, not by the number of incorrect bits.

Characteristics

- Most dangerous type of error
- Common in:
  - Network transmission
  - Wireless communication
  - Optical fiber systems
- Best detected using **Cyclic Redundancy Check (CRC)**



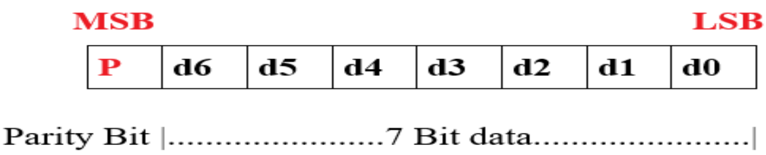
Error Detection Techniques

Error detection techniques **identify the presence of an error** but **do not correct it**.

a. **Parity bit Check-**
b. **Check Sum**
c. **Cyclic Redundancy Check**

a. Parity bit Check:

- Parity is a simple error-detection mechanism used in digital communication and computer memory systems to ensure that data is transmitted or stored **without errors**.
- A **parity bit** is an extra bit added to a group of data bits to make the total number of 1s either even or odd. Parity does not correct errors; it only detects single-bit errors in transmitted or stored **data.**



Or

MSB							LSB
D6	D5	D4	D3	D2	D1	D0	P
………………………. . 7 Bit Data ………………………..							Parity Bit

Types of Parity

- **Even parity**
- **Odd parity**

1. Even Parity

Even parity ensures that the **total number of 1s** in the data word, including the parity bit, is **even**. If the number of 1s in the data is already even, the parity bit is set to 0. If the number of 1s is odd, the parity bit is set to 1 to make the total count even.

Example:
Suppose we want to transmit the 7-bit data 1011001.

1	**0**	**1**	**1**	**0**	**0**	**1**	**X**
7 bit Data							Parity Bit

So total no. of 1's in data bit = 4

So for even parity, the parity bit should be equal to 0

1	**0**	**1**	**1**	**0**	**0**	**1**	**0**
7 bit Data							Parity Bit

Example:2)
Using the same 7-bit data 1011011:

1	**0**	**1**	**1**	**0**	**1**	**1**	**X**
7 bit Data							Parity Bit

So total no. of 1's in data bit = 5

So for even parity, the parity bit should be equal to **1**

1	**0**	**1**	**1**	**0**	**1**	**1**	**1**

7 bit Data	Parity Bit

Even parity is widely used in **RAM memory systems, serial communication**, and **data storage** for simple error detection.

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2. Odd Parity

Odd parity ensures that the **total number of 1s** in the data word, including the parity bit, is **odd**.
If the number of 1s in the data is already odd, the parity bit is set to 0.
If the number of 1s is even, the parity bit is set to 1 to make the total count odd.

Example:1)
Using the same 7-bit data 1011001:

1	0	1	1	0	0	1	X
7 bit Data							Parity Bit

So total no. of 1's in data bit = 4

So for odd parity, the parity bit should be equal to **1**

1	0	1	1	0	0	1	1
7 bit Data							Parity Bit

Example:2)
Using the same 7-bit data 1011011:

1	0	1	1	0	1	1	X
7 bit Data							Parity Bit

So total no. of 1's in data bit = 5

So for odd parity, the parity bit should be equal to **0**

1	0	1	1	0	1	1	0
7 bit Data							Parity Bit

Odd parity is often used in **telecommunication systems** and **serial ports**, where maintaining an odd number of 1s helps the receiver detect single-bit errors.

Error Correction Techniques

Error correction techniques are used in digital communication and computer memory systems to **not only detect errors** but also **correct them** without needing retransmission.

Hamming Code (Error Detection and Correction)

Introduction

- Hamming Code is a widely used **error detection and correction technique** in digital communication systems.
- It was developed by **Richard Hamming** to improve the reliability of data transmission over noisy channels.
- In any communication system, data may get corrupted due to noise. Hamming code helps in **detecting and correcting such errors automatically** without retransmission.

Definition

Hamming Code is a method of encoding data by adding **redundant bits (parity bits)** to the original data bits in such a way that **single-bit errors can be detected and corrected**.

Basic Concept

- Data bits are combined with **parity bits**
- Parity bits are placed at positions that are **powers of 2**
- Each parity bit checks a specific group of bits
- The combination of parity results helps locate errors

Position of Bits

In Hamming Code:

- Parity bits are placed at positions:

$2^0=1$, $2^1=2$, $2^2=4$, $2^3=8$, $2^4=16$ …..

1, 2, 4, 8, 16, ...

Calculate value of parity bits :

Remaining positions are filled with data bits

- Calculate value of parity bits :
- Each parity bit checks specific positions.
- 1. Check the number where position of 1 is at last:

- P1 → checks positions: 1, 3, 5, 7
- 2. Check the number where position of 1 is at middle
- P2 → checks positions: 2, 3, 6, 7
- 3. Check the number where position of 1 is at first
- P3→ checks positions: 4, 5, 6, 7

7	6	5	4	3	2	1
D4	**D3**	**D2**	**P3**	**D1**	**P2**	**P1**
111	**110**	**101**	**100**	**011**	**010**	**001**

What is Cache Memory?

Cache memory is a small, high-speed memory located close to the CPU that stores copies of frequently accessed data and instructions from main memory (RAM), reducing access time and improving overall system performance.
Cache memory increases the access speed of the CPU.

It is not a technique but a memory unit, i.e. a storage device.

Why Cache Memory is Needed?

- CPU speed is **much faster** than RAM.
- Without cache, CPU has to wait for data → slows down processing.
- Cache stores **recently used or frequently used data** → faster access.

In cache memory, recently used data is copied.

Whenever the program is ready to be executed, it is fetched from the main memory and then copied to the cache memory. But, if its copy is already present in the cache memory, then the program is directly executed.

Word Transfer

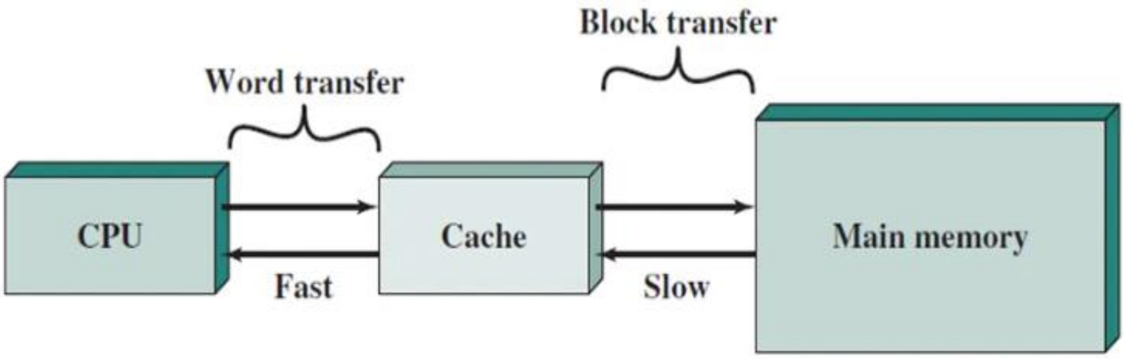
- Transfer of a **single word** (small unit of data, e.g., 1, 2, 4, or 8 bytes) at a time.
- **Where used:** Between **CPU and Cache**. CPU usually needs small, specific data items for immediate processing.

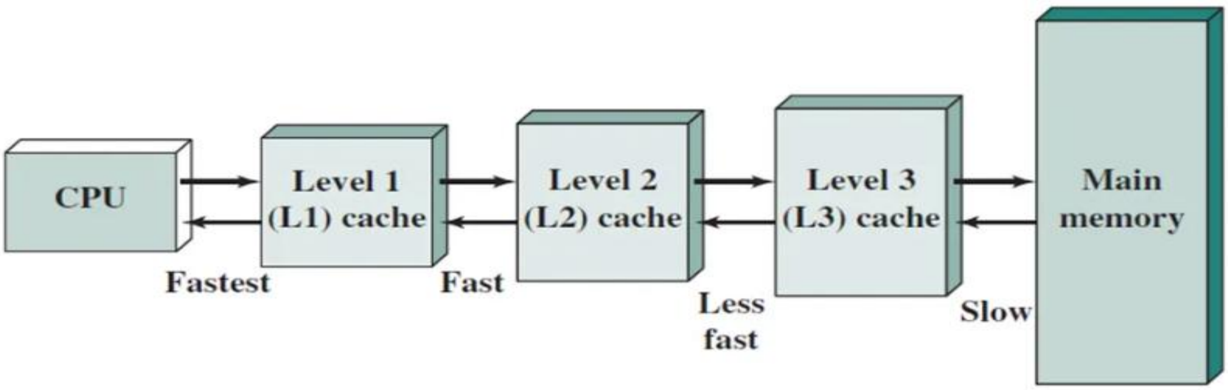
Block Transfer

- Transfer of a **group of words (a block or cache line)** at a time.
- **Where used:** Between **Cache and Main Memory**.

- **Speed: Slower than word transfer**, but efficient for large data movement. Improves performance by bringing a chunk of data that is likely to be used soon.

- If the processor finds that the memory location is in the cache, a **Cache Hit** has occurred and data is read from the cache.
- If the processor does not find the memory location in the cache, a **cache miss** has occurred. For a cache miss, the cache allocates a new entry and copies in data from the main memory, then the request is fulfilled from the contents of the cache.





L1, L2, and L3 cache memory are different levels of cache arranged in a hierarchy to improve the speed of data access for the CPU.

L1 Cache (Level 1)

- **L1 cache** is the first and fastest level of cache memory located directly inside each CPU core.
- It has a very small size, typically ranging from 16 KB to 128 KB, but provides extremely fast access with very low latency.
- L1 cache is usually divided into two parts: instruction cache (I-cache), which stores instructions, and data cache (D-cache), which stores data required for processing.

- Whenever the CPU needs any data or instruction, it first checks the L1 cache. Due to its high speed, it significantly reduces the execution time of programs, but its limited size restricts the amount of data it can store.

L2 cache (Level 2)

- It is the second level of cache memory that comes into use when the required data is not found in L1 cache.
- It is larger in size, typically between 256 KB and 8 MB, but slower than L1 cache.
- It stores less frequently used data compared to L1 but still provides faster access than main memory. L2 cache acts as a bridge between L1 cache and RAM, reducing the number of accesses to slower main memory and thereby improving system performance.

L3 Cache (Level 3)

- **L3 cache** is the third level of cache memory and is larger but slower than both L1 and L2 caches. Its size usually ranges from 4 MB to 64 MB or more.
- When data is not found in L2 cache, the CPU checks L3 cache before accessing the main memory.
- Although it has higher latency, L3 cache plays a crucial role in reducing the load on RAM and enhancing overall system performance.

Advantages of Cache Memory

- **Faster Data Access:** Cache memory offers faster access to data as compared to RAM, and this minimises the time that the CPU has to take to look for data that is frequently used.
- **Improved CPU Efficiency:** Thus, thrashing results from constantly having to access main memory to wait for the information, rather than the CPU efficiently processing instructions.
- **Reduces Bottlenecks:** It assists in getting rid of restrictions that are occasioned by slower memory access speeds, especially where there are numerous tasks that involve usage of many resources.

Disadvantages of Cache Memory

- **Limited Size:** Cache memory is, in fact, a very small type of memory when compared to RAM, and its capacity is not very large.
- **Expensive:** This is because it has a much higher access rate than RAM, as it also resides closer to the CPU and is more costly to manufacture.
- **Complex Management:** Cache data is crucial in increasing system efficiency since it is crucial to control the data contained in the cache as may be deemed appropriate by the algorithms used to manage this data and make certain that only recent data is available in the cache, which makes the system complex.

What is Virtual Memory?

Virtual Memory increases the capacity of main memory. Virtual memory is not a storage unit, it's a technique.

Virtual Memory is a memory management technique that allows a computer to **execute large programs even if the physical RAM is limited**. It gives the illusion of a **very large memory** by using secondary storage (hard disk).

*When RAM is full, the computer uses some space from the **hard disk** like extra RAM.*

Application Start-up (Active RAM)

When you launch applications like a web browser, video editor, and your favourite game, their essential data and instructions are loaded into the super-fast physical **RAM**. This allows for immediate and smooth interaction.

RAM Overload (Memory Full)

As you open more demanding programs or many browser tabs, your computer's physical **RAM capacity might become exhausted**. Without virtual memory, this would often lead to application crashes or system slowdowns.

Data Swapping (Move to Drive)

The operating system then intelligently identifies data in RAM that hasn't been accessed recently (e.g., a minimized application). It moves this data to a designated section on your slower **hard drive** (called a "page file" or "swap space"), freeing up RAM for your active tasks.

Step-by-Step Working-

1. Program Starts

- When you open a program, it is **loaded into RAM**
- Because CPU can only work with data in RAM

2. RAM Becomes Full

- If you open many apps → RAM gets full
- Now system needs extra space

3. Move Data to Disk

- Some **less-used data/programs** are moved from RAM to **hard disk**
- This special disk area is called:
  ☞ **Swap Space / Page File**

4. Program Needs Data Again

- If that moved data is needed again:
  - It is brought back from disk to RAM
  - And some other data may be sent to disk

5. This Process Continues

- Data keeps moving between:
  ☞ RAM ↔ Disk
- So system keeps running smoothly

When is Paging done?

Paging happens whenever a program needs data that is not in RAM

✓ Situations:

- When you open a program → it is divided into pages
- When required page is **not in RAM** → it is brought from disk
- This situation is called:
  ☞ **Page Fault**

So, paging is done:
Whenever needed during program execution (continuously)

When is Swapping done?

Swapping happens when RAM is full and a new process needs memory

✓ Situations:

- Too many programs are running
- RAM is completely occupied
- OS removes a **whole process** from RAM → sends it to disk
- Loads another process into RAM

So, swapping is done:
When RAM is full and space is needed for another process

Advantages of Virtual Memory

1. Run Large Programs
You can run programs bigger than RAM size
No need for large physical memory

2. Multitasking
Many programs can run at the same time
Open browser, editor, apps together

3. Better Memory Usage
Only required data is kept in RAM
Unused data goes to disk → saves space

4. Cost Effective
Less RAM needed, Saves hardware cost

5. Smooth Working
System does not stop even if RAM is full.
Uses disk as backup memory

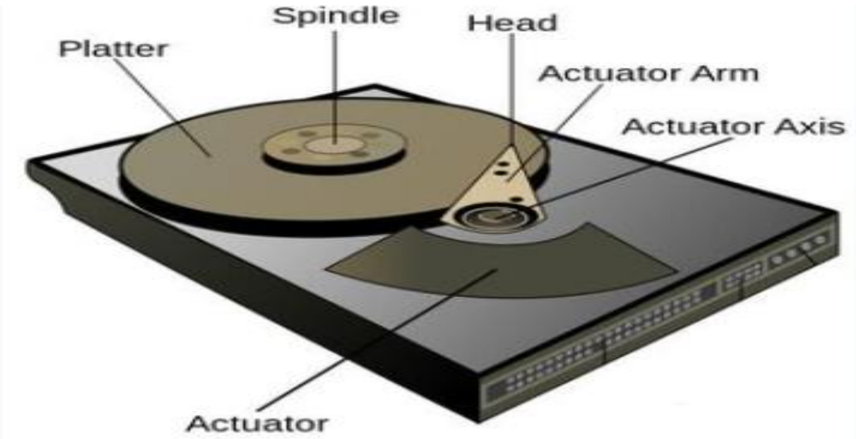
Disadvantages of Virtual Memory

- **Slower Performance:** Whenever information is stored on the hard disk (swap space), it is slower to access compared to RAM.
- **Hard Disk Wear and Tear:** Virtual memory when used excessively can put on more pressure on the hard drive and wear it out more frequently.
- **Requires Management:** It is more complex to manage than RAM like page swapping which takes time if well managed.

External Memory

Magnetic Disk- A **Magnetic Disk** is a secondary storage device used to store data **permanently**. It stores data in the form of **magnetic patterns** on rotating disks called **platters**.

☞ Example: Hard Disk Drive (HDD)



Key Components & Architecture and Architecture-

Platters -> Surface -> Track -> Sector -> Data

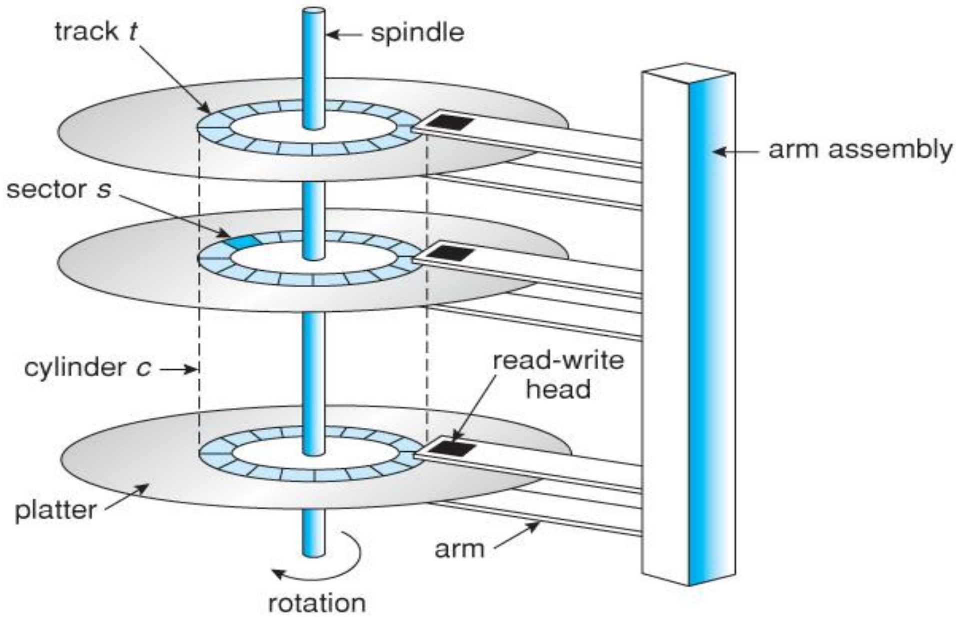
1. **Platters:** The entire disk is divided into platters. Rigid circular disks (often aluminium or glass) coated with magnetic material. Each platter has 2 surfaces- upper and lower and both the surfaces are used to store the data.
2. **Spindle:** A central shaft that holds the platters and spins them at high speeds. These platters are connected with spindle. Through the spindle platers are move/rotate clockwise or anti-clock wise direction means spindle will move or rotate all platters will rotate.
3. **Read/Write Heads:** Each surface has its own read / write head. These read / write heads are connected with arm assembly or actuator arm.

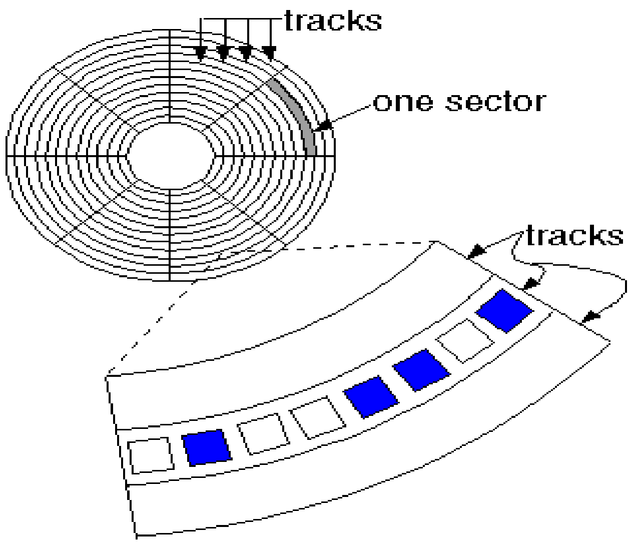
   The possible movement of read / write head is forward and backword.

4. **Track & sectors:** Each working surface is divided into a number of concentric rings called **tracks. No. of tracks is on upper & lower surface are equal.**

   These tracks are further divided into **sectors** which are the smallest divisions in the disk.

5. **A cylinder** is formed by combining the tracks at a given radius of a disk pack.





What is Optical Memory?
Optical memory is a form of electronic storage that uses a laser beam to store and retrieve data. It was developed by Philips and Sony and released in 1982 in the fourth generation of computers. These memories use light beams for their operations and require optical drives for their operations. These memories are used for storing audio/video, backup and caring for data. Read/write speed is slower compared to hard disk and flash memory. Examples of optical memories are Compact Disk (CD), Digital Versatile Disk (DVD), Blu-ray Disc.

Optical Memory

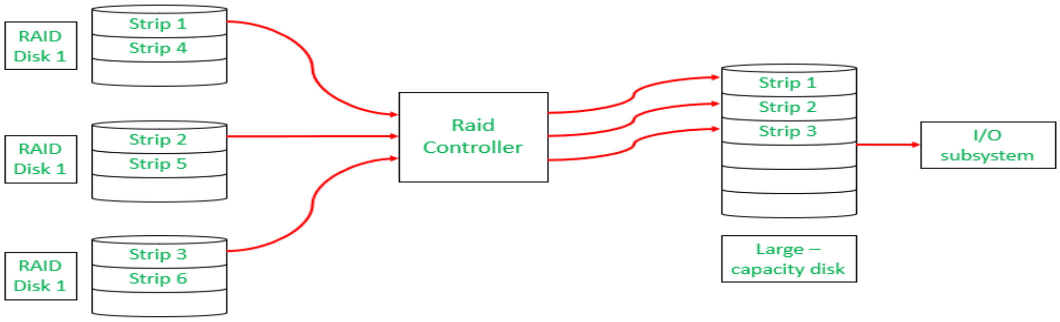
What is Compact Disk (CD) Optical Memory?
It stores data and it has circular plastic, a single side of the plastic is coated by aluminium alloy which stores data. It is protected by an additional thin plastic covering. CD requires a CD drive for its operation. A CD could store much more data than a personal computer hard drive. The CD has storage typically up to 700 MB.

RAID Memory (redundant array of independent disks)

RAID (redundant array of independent disks) is a way of storing the same data in different places on multiple hard disks or solid-state drives (SSDs) to protect data in the case of a drive failure.

What Is RAID Controller?

The central component in a RAID setup is a **RAID controller**. It manages data movement and abstracts the RAID array to the operating system as a single logical unit. The controller handles data distribution, recovery, and redundancy depending on the rules defined at the RAID level.

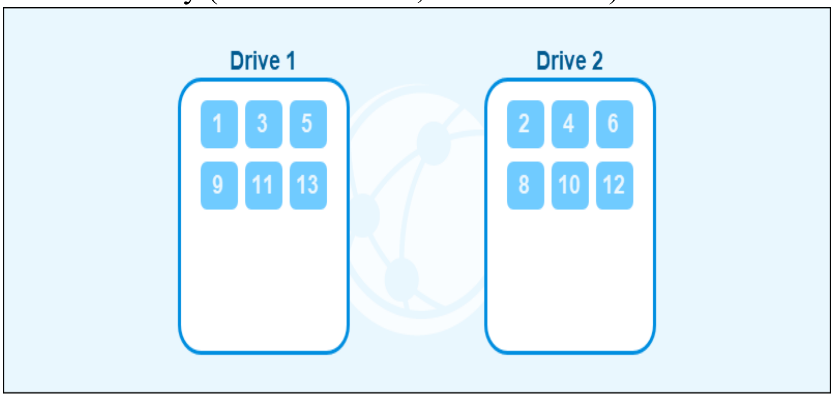


RAID Storage Techniques

RAID uses three main storage techniques to manage data storage and protect data on multiple drives.

1) RAID Striping

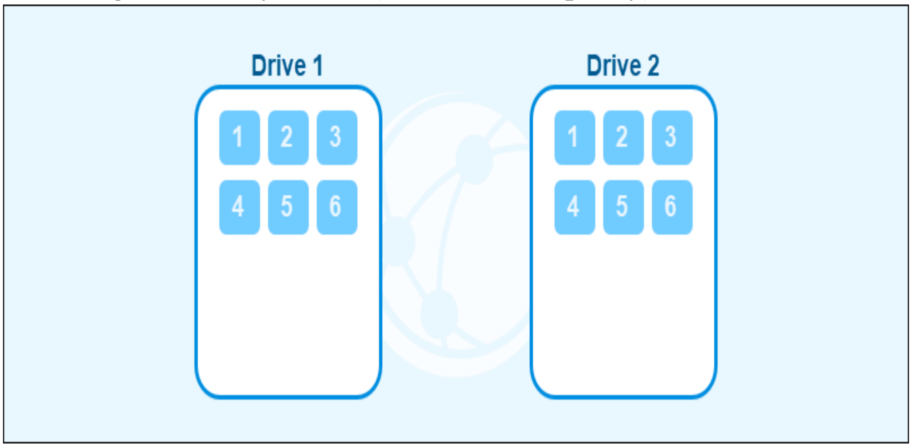
- Data is split into blocks and spread across multiple disks.
- Improves performance because multiple disks read/write simultaneously.
- No redundancy (if one disk fails, all data is lost).



2) RAID Mirroring

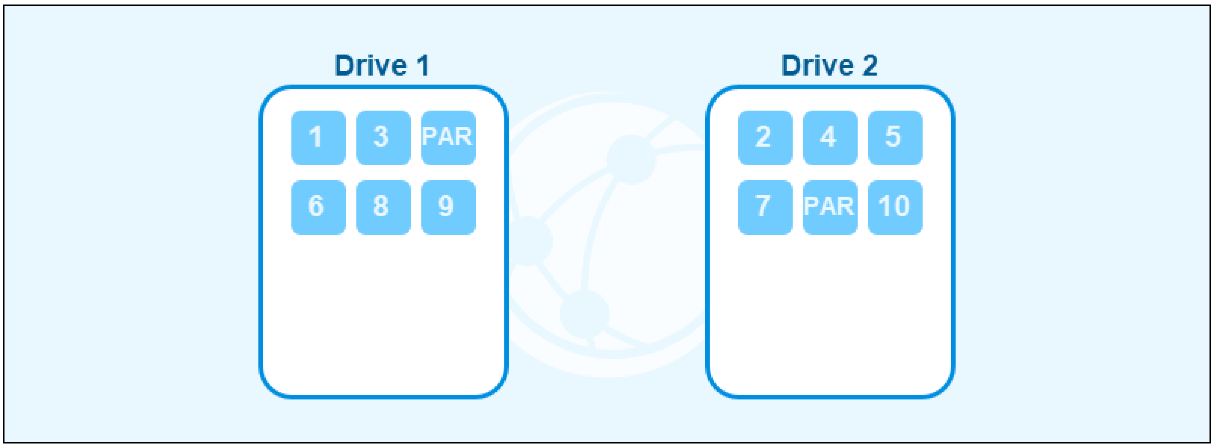
- Data is duplicated exactly on two or more disks.
- Provides high reliability.

- ✓ If one disk fails, data is still available on the other.
- ✕ Storage efficiency is low (50% usable capacity).



3) RAID Parity

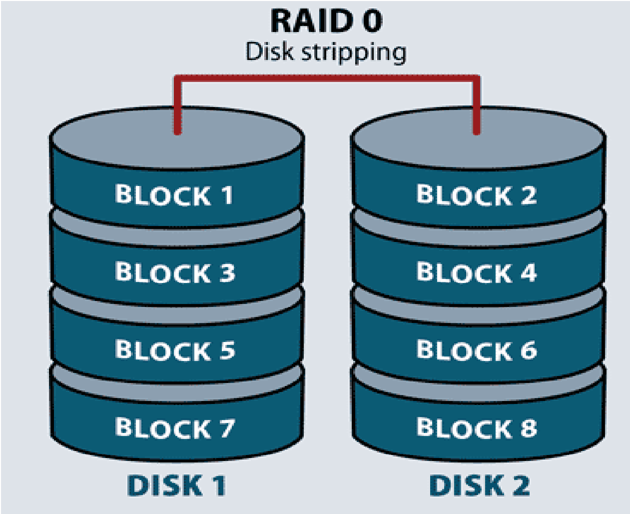
- Extra data (parity) is stored to allow recovery if a disk fails.
- Parity is calculated using a **XOR operation**.
- More storage-efficient than mirroring.



Different RAID levels

1) RAID 0 (Striping)

- Requiring a minimum of two disks, RAID 0 splits files and stripes the data across two disks or more, treating the striped disks as a single partition.
- Because striping technology distributes data across multiple disks, reading one file requires reading multiple disks.
- RAID 0 does not provide redundancy or fault tolerance. Since it treats multiple disks as a single partition, if even one drive fails, the striped file is unreadable.



Advantages of RAID 0

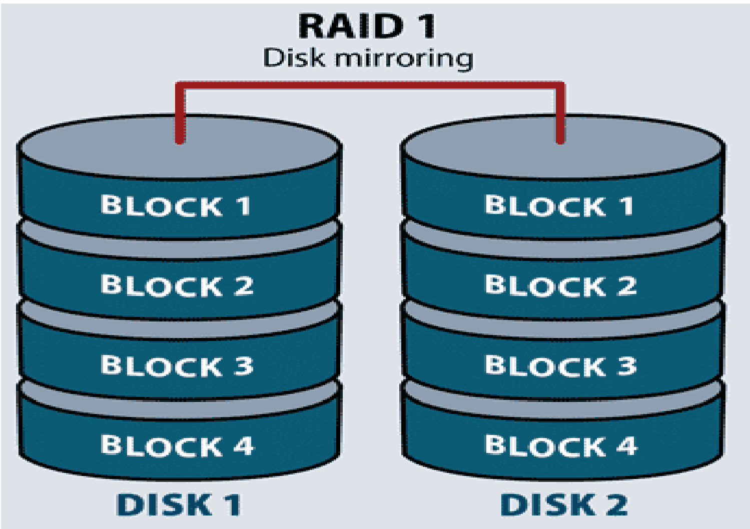
- RAID 0 offers great performance, both in read and write operations.
- All storage capacity is used.
- The technology is easy to implement.

Disadvantages of RAID 0

- RAID 0 is not fault-tolerant. If one drive fails, all data in the RAID 0 array are lost. It should not be used for mission-critical systems.

2) RAID 1- Mirroring

- Data are stored twice by writing them to both the data drive (or set of data drives) and a mirror drive (or set of drives).
- If a drive fails, the controller uses either the data drive or the mirror drive for data recovery and continuous operation

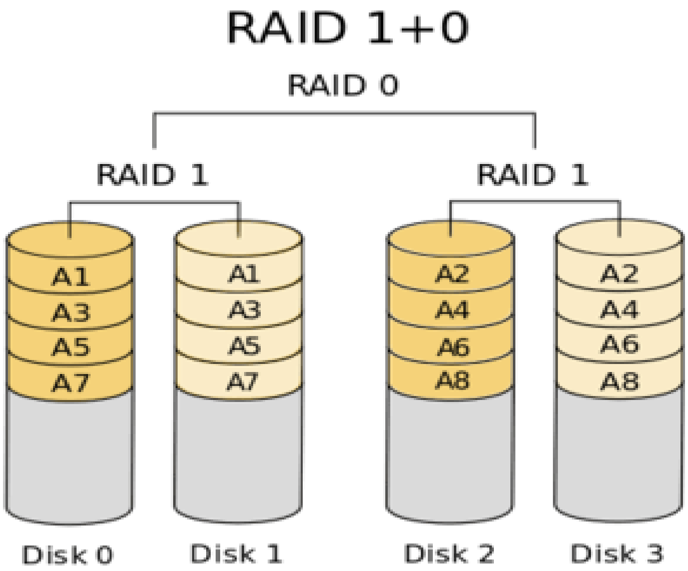


3) RAID 10: Disk mirroring and striping

• It is also called RAID 1+0, is a nested RAID level that combines disk mirroring and striping.

• Data is spread across two or more drives, and multiple read/write heads on the drives can access portions of the data simultaneously, resulting in faster processing.

• Because it uses RAID 1, RAID 10 data is fully protected. If a disk within the set fails or becomes unavailable, the mirror copy can take over.



4) RAID 3: Parity disk

• RAID 3 uses a parity disk to store the parity information generated by a RAID controller on a separate disk from the actual data disks instead of striping it with the data.

• RAID 3 requires a minimum of three physical disks.